

Snippet to take a look at commodities data availability through yfinance

```
In [33]: # Import packages

import pandas as pd
import numpy as np
import yfinance as yf
from pathlib import Path
import matplotlib.pyplot as plt
```

```
In [35]: # Output directory

OUTPUT_DIR = Path(
    r"C:\Users\pcarg\OneDrive\Υπολογιστής\MFE UCLA\Projects\Commodities\Data"
)

OUTPUT_DIR.mkdir(
    parents=True,
    exist_ok=True
)
```

```
In [3]: # Commodities universe

commodities = {

    # =====
    # ENERGY
    # =====

    "WTI Crude": "CL=F",
    "Brent": "BZ=F",
    "Natural Gas": "NG=F",
    "RBOB": "RB=F",
    "HeatingOil": "HO=F",

    # =====
    # PRECIOUS METALS
    # =====

    "Gold": "GC=F",
    "Silver": "SI=F",
    "Platinum": "PL=F",
    "Palladium": "PA=F",

    # =====
    # INDUSTRIAL METALS
    # =====

    "Copper": "HG=F",
```

```

# =====
# GRAINS
# =====

"Corn": "ZC=F",
"Soybeans": "ZS=F",
"Wheat": "ZW=F",
"KC_Wheat": "KE=F",
"Rice": "ZR=F",
"Oats": "ZO=F",

# =====
# SOFTS
# =====

"Coffee": "KC=F",
"Sugar": "SB=F",
"Cocoa": "CC=F",
"Cotton": "CT=F",
"OrangeJuice": "OJ=F",

# =====
# LIVESTOCK
# =====

"LiveCattle": "LE=F",
"LeanHogs": "HE=F",
"FeederCattle": "GF=F"
}

```

In [5]: # *Other futures universe*

```

futures = {

# =====
# EQUITY INDEX
# =====

"SP500": "ES=F",
"Nasdaq100": "NQ=F",
"Dow": "YM=F",
"Russell2000": "RTY=F",

# =====
# RATES
# =====

"US_2Y": "ZT=F",
"US_5Y": "ZF=F",
"US_10Y": "ZN=F",
"US_30Y": "ZB=F",
"UltraBond": "UB=F",

# =====
# FX
# =====

```

```

    "EURUSD": "6E=F",
    "JPYUSD": "6J=F",
    "GBPUSD": "6B=F",
    "AUDUSD": "6A=F",
    "CADUSD": "6C=F"
}

```

In [7]: *# Combine into master*

```

universe = {
    **commodities,
    **futures
}

print(f"Number of assets: {len(universe)}")

```

Number of assets: 38

In [9]: *# Audit yahoo coverage*

```

audit = []

for asset, ticker in universe.items():

    try:

        df = yf.download(
            ticker,
            period="max",
            auto_adjust=False,
            progress=False
        )

        audit.append({
            "Asset": asset,
            "Ticker": ticker,
            "Rows": len(df),
            "Start": df.index.min(),
            "End": df.index.max()
        })

    except Exception:

        audit.append({
            "Asset": asset,
            "Ticker": ticker,
            "Rows": 0,
            "Start": None,
            "End": None
        })

audit = pd.DataFrame(audit)

audit.sort_values(
    "Rows",

```

```
ascending=False  
)
```

Out[9]:

	Asset	Ticker	Rows	Start	End
14	Rice	ZR=F	6700	1999-09-14	2026-06-07
19	Cotton	CT=F	6627	2000-01-03	2026-06-05
18	Cocoa	CC=F	6627	2000-01-03	2026-06-05
16	Coffee	KC=F	6625	2000-01-03	2026-06-05
17	Sugar	SB=F	6588	2000-03-01	2026-06-05
15	Oats	ZO=F	6557	1999-09-14	2026-06-07
28	US_2Y	ZT=F	6522	2000-06-02	2026-06-07
37	CADUSD	6C=F	6516	2000-05-23	2026-06-07
8	Palladium	PA=F	6505	1998-09-28	2026-06-07
33	EURUSD	6E=F	6496	2000-09-12	2026-06-07
24	SP500	ES=F	6494	2000-09-18	2026-06-07
25	Nasdaq100	NQ=F	6494	2000-09-18	2026-06-07
7	Platinum	PL=F	6494	1997-10-29	2026-06-07
12	Wheat	ZW=F	6490	2000-07-17	2026-06-07
10	Corn	ZC=F	6478	2000-07-17	2026-06-07
0	WTI Crude	CL=F	6475	2000-08-23	2026-06-07
2	Natural Gas	NG=F	6472	2000-08-30	2026-06-07
9	Copper	HG=F	6471	2000-08-30	2026-06-07
11	Soybeans	ZS=F	6470	2000-09-15	2026-06-07
4	HeatingOil	HO=F	6469	2000-09-01	2026-06-07
6	Silver	SI=F	6468	2000-08-30	2026-06-07
29	US_5Y	ZF=F	6468	2000-09-21	2026-06-07
13	KC_Wheat	KE=F	6467	2000-09-21	2026-06-07
5	Gold	GC=F	6466	2000-08-30	2026-06-07
31	US_30Y	ZB=F	6462	2000-09-21	2026-06-07
30	US_10Y	ZN=F	6456	2000-09-21	2026-06-07
3	RBOB	RB=F	6430	2000-11-01	2026-06-07
34	JPYUSD	6J=F	6419	2000-09-13	2026-06-07
35	GBPUSD	6B=F	6410	2000-10-05	2026-06-07
22	LeanHogs	HE=F	6400	2000-12-15	2026-06-05

	Asset	Ticker	Rows	Start	End
36	AUDUSD	6A=F	6399	2000-11-08	2026-06-07
21	LiveCattle	LE=F	6332	2001-03-01	2026-06-05
23	FeederCattle	GF=F	6302	2001-04-03	2026-06-05
20	OrangeJuice	OJ=F	6203	2001-09-17	2026-06-05
26	Dow	YM=F	6085	2002-04-05	2026-06-07
1	Brent	BZ=F	4693	2007-07-30	2026-06-07
32	UltraBond	UB=F	4126	2010-01-11	2026-06-07
27	Russell2000	RTY=F	2244	2017-07-10	2026-06-07

In [11]: *# Inspect failed assets*

```
audit[audit["Rows"] == 0]
```

Out[11]:

Asset	Ticker	Rows	Start	End
-------	--------	------	-------	-----

In [13]: *# Generate container dataframes*

```
open_prices = pd.DataFrame()
high_prices = pd.DataFrame()
low_prices = pd.DataFrame()
close_prices = pd.DataFrame()
adj_close_prices = pd.DataFrame()
volumes = pd.DataFrame()

metadata = []
```

In [15]: *# Download production dataset*

```
for asset, ticker in universe.items():

    print(f"Downloading {asset} ({ticker})")

    try:

        df = yf.download(
            ticker,
            period="max",
            auto_adjust=False,
            progress=False
        )

        if len(df) == 0:
            continue

        open_prices[asset] = df["Open"]
        high_prices[asset] = df["High"]
```

```
low_prices[asset] = df["Low"]
close_prices[asset] = df["Close"]
adj_close_prices[asset] = df["Adj Close"]
volumes[asset] = df["Volume"]

metadata.append({
    "Asset": asset,
    "Ticker": ticker,
    "Rows": len(df),
    "Start": df.index.min(),
    "End": df.index.max()
})

except Exception as e:

    print(
        f"FAILED -> {asset} ({ticker})"
    )

    print(e)
```

```
Downloading WTI Crude (CL=F)
Downloading Brent (BZ=F)
Downloading Natural Gas (NG=F)
Downloading RBOB (RB=F)
Downloading HeatingOil (HO=F)
Downloading Gold (GC=F)
Downloading Silver (SI=F)
Downloading Platinum (PL=F)
Downloading Palladium (PA=F)
Downloading Copper (HG=F)
Downloading Corn (ZC=F)
Downloading Soybeans (ZS=F)
Downloading Wheat (ZW=F)
Downloading KC_Wheat (KE=F)
Downloading Rice (ZR=F)
Downloading Oats (ZO=F)
Downloading Coffee (KC=F)
Downloading Sugar (SB=F)
Downloading Cocoa (CC=F)
Downloading Cotton (CT=F)
Downloading OrangeJuice (OJ=F)
Downloading LiveCattle (LE=F)
Downloading LeanHogs (HE=F)
Downloading FeederCattle (GF=F)
Downloading SP500 (ES=F)
Downloading Nasdaq100 (NQ=F)
Downloading Dow (YM=F)
Downloading Russell2000 (RTY=F)
Downloading US_2Y (ZT=F)
Downloading US_5Y (ZF=F)
Downloading US_10Y (ZN=F)
Downloading US_30Y (ZB=F)
Downloading UltraBond (UB=F)
Downloading EURUSD (6E=F)
Downloading JPYUSD (6J=F)
Downloading GBPUSD (6B=F)
Downloading AUDUSD (6A=F)
Downloading CADUSD (6C=F)
```

In [17]: *# Sort index*

```
datasets = [
    open_prices,
    high_prices,
    low_prices,
    close_prices,
    adj_close_prices,
    volumes
]

for df in datasets:

    df.sort_index(inplace=True)

    df.index.name = "Date"
```



```
In [19]: # Metadata

metadata = pd.DataFrame(metadata)

metadata.sort_values(
    "Rows",
    ascending=False
)
```

Out[19]:

	Asset	Ticker	Rows	Start	End
14	Rice	ZR=F	6700	1999-09-14	2026-06-07
19	Cotton	CT=F	6627	2000-01-03	2026-06-05
18	Cocoa	CC=F	6627	2000-01-03	2026-06-05
16	Coffee	KC=F	6625	2000-01-03	2026-06-05
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7	Platinum	PL=F	6494	1997-10-29	2026-06-07
12	Wheat	ZW=F	6490	2000-07-17	2026-06-07
10	Corn	ZC=F	6478	2000-07-17	2026-06-07
0	WTI Crude	CL=F	6475	2000-08-23	2026-06-07
2	Natural Gas	NG=F	6472	2000-08-30	2026-06-07
9	Copper	HG=F	6471	2000-08-30	2026-06-07
11	Soybeans	ZS=F	6470	2000-09-15	2026-06-07
4	HeatingOil	HO=F	6469	2000-09-01	2026-06-07
6	Silver	SI=F	6468	2000-08-30	2026-06-07
29	US_5Y	ZF=F	6468	2000-09-21	2026-06-07
13	KC_Wheat	KE=F	6467	2000-09-21	2026-06-07
5	Gold	GC=F	6466	2000-08-30	2026-06-07
31	US_30Y	ZB=F	6462	2000-09-21	2026-06-07
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34	JPYUSD	6J=F	6419	2000-09-13	2026-06-07
35	GBPUSD	6B=F	6410	2000-10-05	2026-06-07
22	LeanHogs	HE=F	6400	2000-12-15	2026-06-05

	Asset	Ticker	Rows	Start	End
36	AUDUSD	6A=F	6399	2000-11-08	2026-06-07
21	LiveCattle	LE=F	6332	2001-03-01	2026-06-05
23	FeederCattle	GF=F	6302	2001-04-03	2026-06-05
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26	Dow	YM=F	6085	2002-04-05	2026-06-07
1	Brent	BZ=F	4693	2007-07-30	2026-06-07
32	UltraBond	UB=F	4126	2010-01-11	2026-06-07
27	Russell2000	RTY=F	2244	2017-07-10	2026-06-07

```
In [21]: # Diagnostics for missing data

missing_summary = pd.DataFrame({
    "Missing Count":
        close_prices.isna().sum(),

    "Missing %":
        100 * close_prices.isna().mean()
})

missing_summary.sort_values(
    "Missing %",
    ascending=False
)
```

Out[21]:

	Missing Count	Missing %
Russell2000	4232	65.359073
UltraBond	2350	36.293436
Brent	1783	27.536680
Platinum	687	10.610039
Palladium	448	6.918919
Dow	402	6.208494
OrangeJuice	274	4.231660
FeederCattle	189	2.918919
Oats	175	2.702703
LiveCattle	158	2.440154
AUDUSD	116	1.791506
GBPUSD	104	1.606178
JPYUSD	96	1.482625
LeanHogs	92	1.420849
RBOB	50	0.772201
Corn	42	0.648649
US_10Y	37	0.571429
Wheat	31	0.478764
US_30Y	31	0.478764
US_2Y	27	0.416988
US_5Y	26	0.401544
Soybeans	24	0.370656
Rice	23	0.355212
KC_Wheat	22	0.339768
SP500	19	0.293436
Nasdaq100	19	0.293436
EURUSD	18	0.277992
CADUSD	18	0.277992
Coffee	13	0.200772
Cotton	11	0.169884

	Missing Count	Missing %
Cocoa	11	0.169884
Sugar	11	0.169884
Gold	10	0.154440
Silver	9	0.138996
Copper	8	0.123552
HeatingOil	8	0.123552
Natural Gas	5	0.077220
WTI Crude	0	0.000000

```
In [23]: # Start dates

start_dates = pd.DataFrame({
    "First Observation":
        close_prices.apply(
            lambda x: x.first_valid_index()
        )
})

start_dates.sort_values(
    "First Observation"
)
```

Out[23]:

First Observation	
WTI Crude	2000-08-23
US_2Y	2000-08-23
Cotton	2000-08-23
Sugar	2000-08-23
Coffee	2000-08-23
Oats	2000-08-23
Rice	2000-08-23
Wheat	2000-08-23
Corn	2000-08-23
Cocoa	2000-08-23
Palladium	2000-08-23
Platinum	2000-08-23
CADUSD	2000-08-23
Silver	2000-08-30
Gold	2000-08-30
Natural Gas	2000-08-30
Copper	2000-08-30
HeatingOil	2000-09-01
EURUSD	2000-09-12
JPYUSD	2000-09-13
Soybeans	2000-09-15
Nasdaq100	2000-09-18
SP500	2000-09-18
US_30Y	2000-09-21
US_10Y	2000-09-21
US_5Y	2000-09-21
KC_Wheat	2000-09-21
GBPUSD	2000-10-05
RBOB	2000-11-01
AUDUSD	2000-11-08

First Observation	
LeanHogs	2000-12-15
LiveCattle	2001-03-01
FeederCattle	2001-04-03
OrangeJuice	2001-09-17
Dow	2002-04-05
Brent	2007-07-30
UltraBond	2010-01-11
Russell2000	2017-07-10

```
In [25]: # End dates

end_dates = pd.DataFrame({
    "Last Observation":
        close_prices.apply(
            lambda x: x.last_valid_index()
        )
})

end_dates.sort_values(
    "Last Observation"
)
```

Out[25]:

Last Observation	
Cocoa	2026-06-05
Sugar	2026-06-05
Coffee	2026-06-05
LiveCattle	2026-06-05
LeanHogs	2026-06-05
FeederCattle	2026-06-05
OrangeJuice	2026-06-05
Cotton	2026-06-05
US_2Y	2026-06-07
US_10Y	2026-06-07
Dow	2026-06-07
Nasdaq100	2026-06-07
SP500	2026-06-07
US_30Y	2026-06-07
UltraBond	2026-06-07
EURUSD	2026-06-07
JPYUSD	2026-06-07
GBPUSD	2026-06-07
US_5Y	2026-06-07
Russell2000	2026-06-07
WTI Crude	2026-06-07
Oats	2026-06-07
Brent	2026-06-07
Natural Gas	2026-06-07
RBOB	2026-06-07
HeatingOil	2026-06-07
Gold	2026-06-07
Silver	2026-06-07
AUDUSD	2026-06-07
Platinum	2026-06-07

Last Observation	
Copper	2026-06-07
Corn	2026-06-07
Soybeans	2026-06-07
Wheat	2026-06-07
KC_Wheat	2026-06-07
Rice	2026-06-07
Palladium	2026-06-07
CADUSD	2026-06-07

```
In [31]: # Heatmap for missingness

plt.figure(figsize=(16,10))

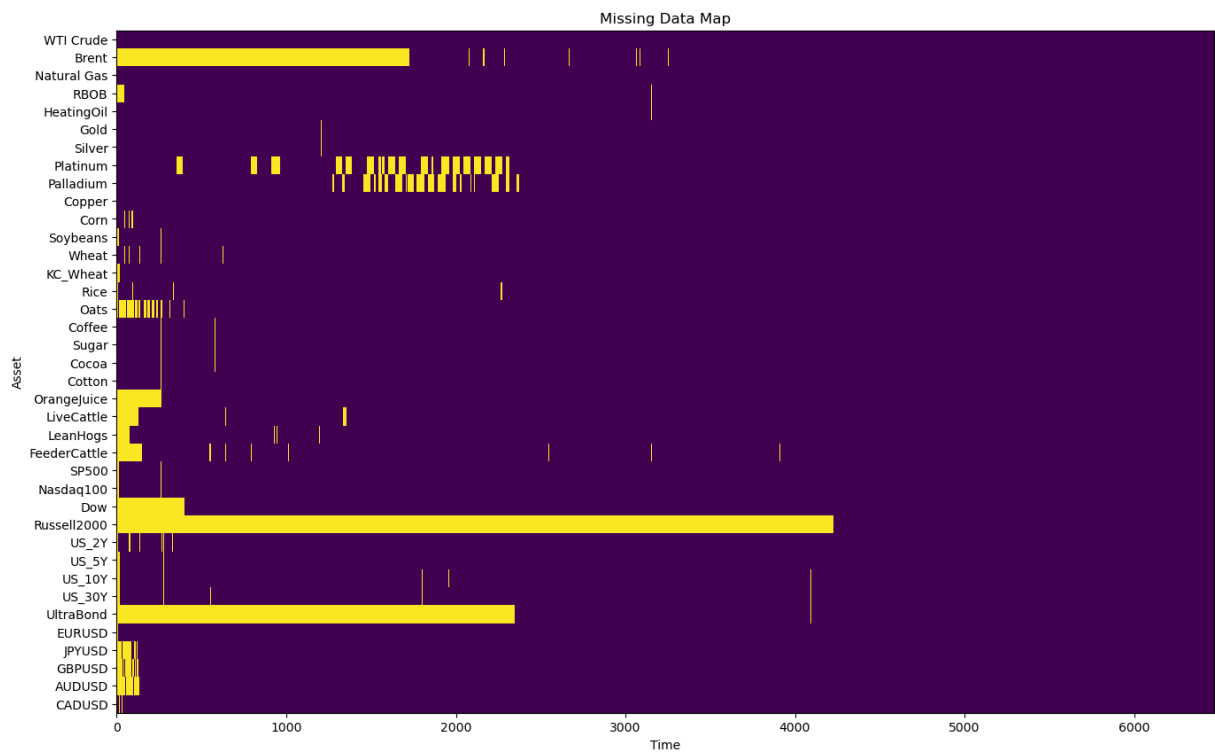
plt.imshow(
    close_prices.isna().T,
    aspect="auto",
    interpolation="none"
)

plt.yticks(
    range(len(close_prices.columns)),
    close_prices.columns
)

plt.xlabel("Time")
plt.ylabel("Asset")

plt.title(
    "Missing Data Map"
)

plt.show()
```



```
In [37]: # Save files

open_prices.to_parquet(
    OUTPUT_DIR / "open.parquet"
)

high_prices.to_parquet(
    OUTPUT_DIR / "high.parquet"
)

low_prices.to_parquet(
    OUTPUT_DIR / "low.parquet"
)

close_prices.to_parquet(
    OUTPUT_DIR / "close.parquet"
)

adj_close_prices.to_parquet(
    OUTPUT_DIR / "adj_close.parquet"
)

volumes.to_parquet(
    OUTPUT_DIR / "volume.parquet"
)

# Metadata

metadata.to_csv(
    OUTPUT_DIR / "metadata.csv",
```

```
    index=False  
)
```

```
In [39]: # Final check  
  
audit.to_csv(  
    OUTPUT_DIR / "audit.csv",  
    index=False  
)  
  
print(f"Files saved to:\n{OUTPUT_DIR}")
```

Files saved to:

C:\Users\pcarg\OneDrive\Υπολογιστής\MFE UCLA\Projects\Commodities\Data